Assignment 4 – Style Transfer

Introduction

The paper *Image Style Using Convolutional Neural Networks* introduced a technique to combine the content of a photograph with the appearance of a work of art. In this exercise, one photo was used as the content target and three distinct works of art were used for style extraction.

Methods

The focus for the experimentation was to analyze the difference between heavier weighting on the lower levels (near the input) and the higher levels (near the output). The expectation from reading the paper is that the lower levels will result in smaller detail transfer like texture, and the higher levels will result in more transfer of larger, more abstract features. Hyperparameters had to be tuned carefully to see the distinction of the respective weighting schemes.

Low Level Heavy Extraction Parameters

Through experimentation with different hyperparameter settings to obtain distinct results for low- and high-level feature extraction, the following settings were found to be effective.

Low level feature extraction favors the first convolutional block and falls sharply to zero by the last block. Alpha and Beta values of 1 and 1e6 were effective.

```
style_weights = {'conv1_1': 1,
                 'conv2_1': 0.75,
                 'conv3_1': 0.25,
                 'conv4_1': 0.1,
                 'conv5_1': 0}

content_weight = 1  # alpha
style_weight = 1e6  # beta
```

Fig. 1. Low Level weighting parameters.

Cosine annealing learning rate scheduling on an ADAM optimizer was chosen so that an aggressive learning rate can be set for the initial iterations, while allowing a smaller learning rate towards the end for stabilization. The range of learning rates was, after experimenting with different rates, set to begin at 0.1 and end at 0.0004.

```
# iteration hyperparameters
steps = 20000

# optimizer
optimizer = optim.Adam([target], lr=0.1)

# learning-rate scheduler
scheduler = torch.optim.lr_scheduler.CosineAnnealingLR(
    optimizer,
    T_max=steps,
    eta_min=1e-4
)
```

Fig. 2. Low level optimization parameters.

High Level Heavy Extraction Parameters

For high level feature extraction, the pattern used for the low level feature extraction was flipped upside down for a distinctive effect. The beta value had to be increased drastically to make the effect dramatic enough for this analysis.

```
style_weights = {'conv1_1': 0,
                 'conv2_1': 0.1,
                 'conv3_1': 0.25,
                 'conv4_1': 0.75,
                 'conv5_1': 1}

content_weight = 1  # alpha
style_weight = 1e12  # beta
```

Fig. 3. High level weighting parameters.

A similar learning rate scheduler was implemented for high level feature extraction. However, for the high-level features, the lower end of the scheduler had to be pulled up to 0.01. This increases the learning rate generally throughout the optimization process.

```python
# iteration hyperparameters
steps = 20000

# optimizer
optimizer = optim.Adam([target], lr=0.1)

# learning-rate scheduler
scheduler = torch.optim.lr_scheduler.CosineAnnealingLR(
    optimizer,
    T_max=steps,
    eta_min=1e-2
)
```

Fig. 4. High level optimization parameters.

A photograph of a well-known building on the Chico State campus was used for the content image because it contains a range of features like a building, foreground grass, background sky, bushes and trees.

Fig. 5. Content Image is a photo from Chico State Campus.

Three paintings of very different texture styles were used for the style transfer images. *Japanese Bridge* and *Woman in the Garden*, by Claude Monet, and *Silver Pavilion – Kyoto* by Toshi Yoshida. These each have very different structure, with *The Japanese Bridge* being very streaky and blurry with large strokes, *Woman in the Garden* having large contrast and a dotted texture, and *Silver Pavilion* being a woodblock print with a unique style of its own.

Results / Analysis

Fig. 6. Style Image 1: *Woman in the Garden*, Claude Monet.

Low level feature transfer resulted in the colors of the style image being applied to the content image. The process seems to have preserved the edges of the content image for the most part, but interestingly the process split the colors in the sky to make it look like there was a smaller cloud.

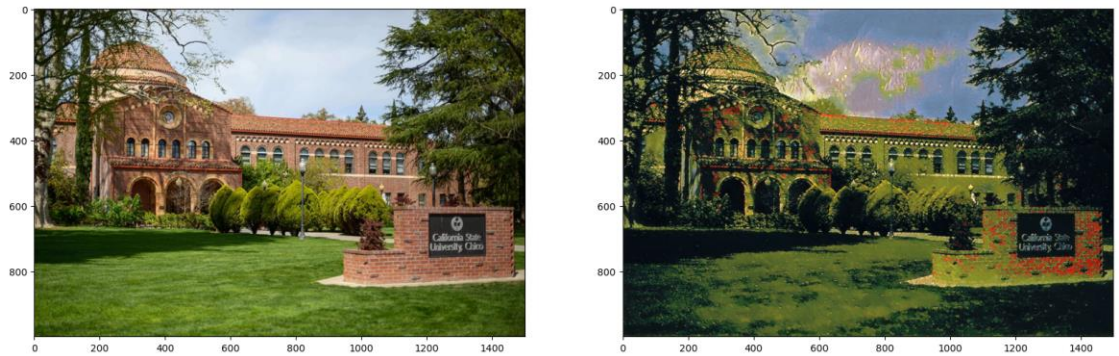

Fig. 7. Low layer heavy, style image 1.

High-Level feature transfer did more to preserve the colors of the content image but introduced shapes from the style image. The speckled paint pattern shows up throughout the generated image, and roses appear in multiple locations.

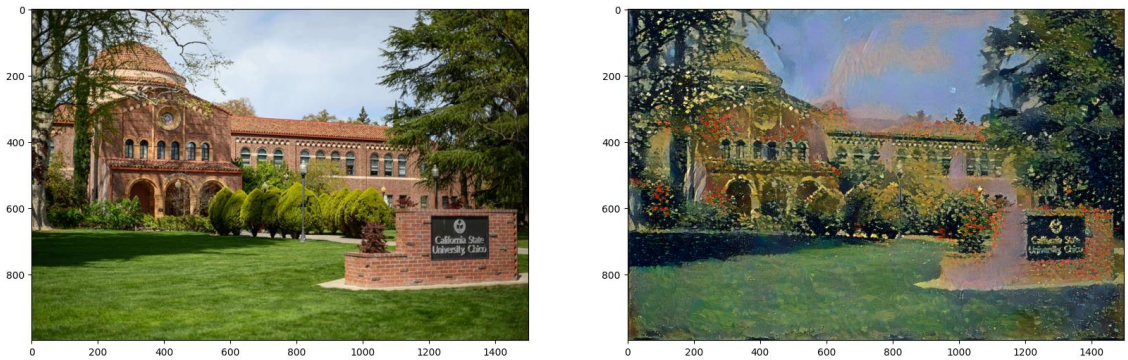

Fig. 8. High level heavy, style image 1.

Fig. 9. Style Image 2: *Silver Pavilion – Kyoto*, Toshi Yoshida.

Low level feature transfer from the woodblock print by Yoshida had a similar effect. The low-level features predominantly selected the closest colors from the content image and applied respective colors from the style image. Some of the smaller texture patterns also show up in the sky and the grass, appearing like paper.

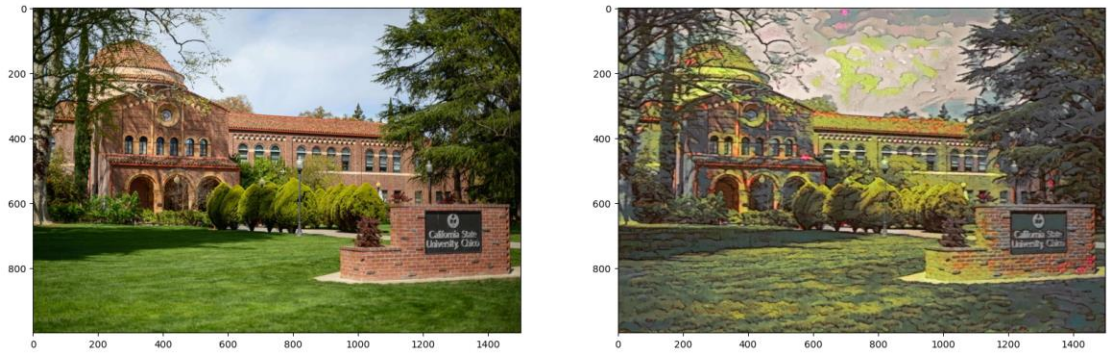

Fig. 5. Low layer heavy, style image 2.

High level feature extraction resulted in a very interesting transformation. The generated image looks like it is a woodblock print or watercolor painting itself. There are paint pooling effects in the bottom corners and strong paper textures. The tree on the right almost completely transformed into a type of tree from the style image. The lines on the building became wavy and watery and the grass took on a complex and large pattern.

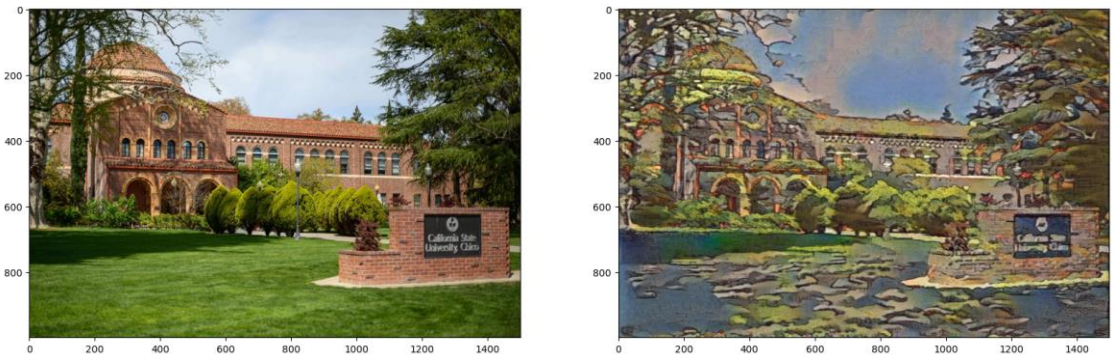

Fig. 6. High layer heavy, style image 2.

Fig. 7. Style Image 3: *Japanese Bridge*, Claude Monet.

Low level style transfer from *Japanese Bridge* by Monet much like the process from the other style images, mostly transferred the color from the style image while preserving the shapes of the content image. It took the prominent brush strokes of the style image and applied them to the generated image but at a small scale, appearing like pencil marks.

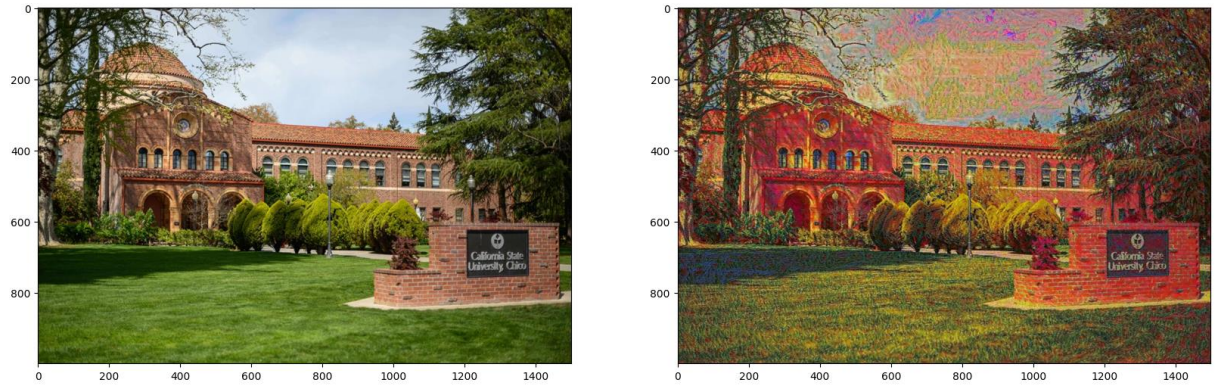

Fig. 8. Low layer heavy, style image 3.

High level feature transfer again resulted in impressive transfer of style with decent preservation of the colors from the content image. The grass and the trees look like they were brushed with astonishing authenticity yet have the same overall form of content image. The brush strokes look much like Monet's but conform to the structure of the photo.

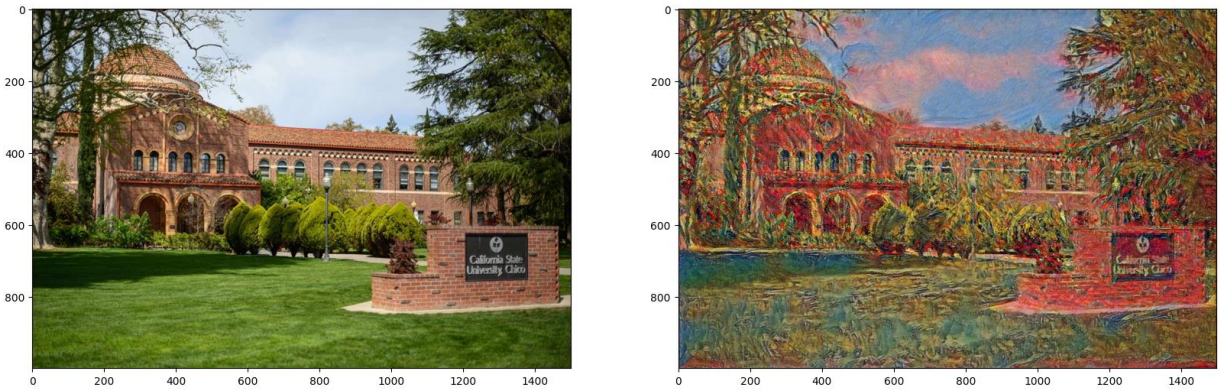

Fig. 9. High layer heavy, style image 3.

Discussion / Conclusion

The goal of analyzing the different effects of low level and high-level feature extraction was successful, for multiple style pieces with distinct textures. Different hyperparameters for effective transfer of high- and low-level features were determined by experimentation. High-level feature transfer required far higher style weight and higher learning rate to be noticeable in the generated image. Generally, low-level feature transfer resulted in the colors from the style image being applied to the content image wherever a most similar color existed, creating a sort of color quantized effect. Small textural qualities were also transferred with low level features. High-level feature transfer did more to preserve the color of the content image while applying larger qualities of the style image, such as brushstroke style and even artistic style. Overall, I'm more impressed with the results of the high-level transfer, particularly with the results from style image 2 (figure 6). The artifacts of the paper and the way the paint blotches while drying looks very interesting and in some way even more real than the actual style image. The way some of the trees completely took on the style of the trees in the style image is remarkable. If I ever was to repeat this process I would start with those hyperparameters I found for the high-level features.

[1] https://www.cvfoundation.org/openaccess/content_cvpr_2016/papers/Gatys_Image_Style_Transfer_CVPR_2016_paper.pdf